

SULIT



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of
Computing

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER II, 2017 / 2018

SUBJECT CODE : SCSV 2113
SUBJECT NAME : HUMAN COMPUTER INTERACTION
SECTION : ALL
TIME : 9.00 AM – 12. 00 NOON
DATE/DAY : 09 / 06 / 2018 (SATURDAY)
VENUES : DEWAN SERBAGUNA KTDI

INSTRUCTIONS :

THIS EXAMINATION BOOK CONSISTS OF TWO (2) PARTS:

- **PART A : 3 STRUCTURED QUESTIONS (33 MARKS)**
- **PART B : 2 LONG QUESTIONS (67 MARKS)**
-

ANSWER ALL QUESTIONS IN THE ANSWER BOOKLET PROVIDED.

(Please Write Your Lecture Name And Section In Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This questions paper consists of **FIVE (5)** printed pages excluding this page.

PART A: STRUCTURED QUESTIONS

[33 Marks]

This part consists of **FIVE short and structured questions**. The marks for each part of the question is as indicated.

1. The following (a) and (b) questions are related to one another.

- a. Siti is a Web designer for IT Solution Company. She has to design a client's website. Before she starts her project, she has to do four basic activities in Interaction Design (ID). What are the **FOUR** basic activities will involve in her project? She needs to identify the practical issues in ID process. Give **THREE** practical issues that Siti needs to consider while she is starting her project.

(7 marks)

- b. Give the **DEFINITION** of quality thresholds. List and explain **THREE** quality thresholds that will help Siti to choose among the alternatives for her project.

(4 marks)

2. The following questions are on Gestalt Principles.

- a. Briefly describe the idea of Gestalt principles.

(1 mark)

- b. For each Figure A.1 (i, ii, iii and iv) below, label the Gestalt principles of visual organisation and *briefly* explain how we (humans) interpret such visuals.

(6 marks)

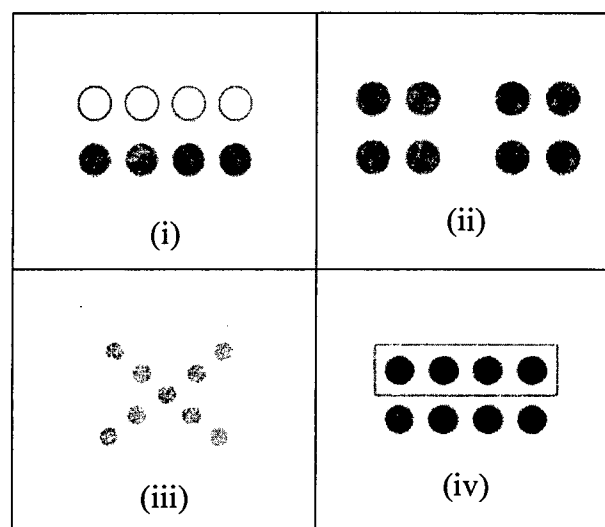


Figure A.1: Illustrations to Gestalt Principles

- c. Discuss how the landing page for an online shopping site (shown in Figure A.2) exploits the Figure and Ground relationship of the Gestalt principle.

(4 marks)

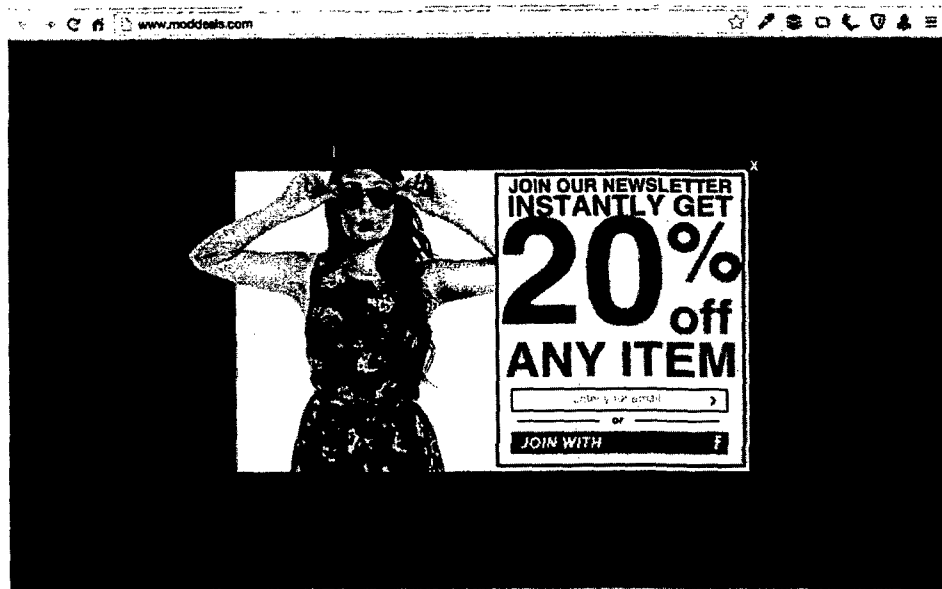


Figure A.2: Landing page to online shopping website

3. By giving an example, describe when you would want to perform an evaluation in controlled settings as opposed to natural settings.

(4 marks)

4. One of the advantages of having the same participants in both conditions (i.e. within subjects) of an experiment is we do not have to go extra length to recruit large number of subjects. But by doing so, *order effect* may take place. By giving an example of 10 participants for condition A and B, explain how we can overcome this.

(4 marks)

5. Evaluation can also be performed without involving users as participants. Succinctly distinguish with one statement the differences between discount evaluation vs. cognitive walkthrough vs. pluralistic walkthrough.

(3 marks)

PART B: LONG QUESTION

[67 Marks]

This part consists of **TWO** long questions. The mark for each part of the question is as indicated.

Question 1

[34 marks]

Massive Open Online Course (MOOC) is an online learning platform aimed at large scale interactive participative learning and open access via the world wide web. MOOC enhanced and fused traditional course materials such as videos, power point slides, readings, and quizzes with interactivity. MOOCs usually comes with interactive user forums and online community for students, professors, and teachers. MOOCs targeted to liberalise education from the classroom settings.

On the other hand, modern classroom settings such as in Figure B.1 allows the professor and students to have face-to-face immediate interactions and feedbacks, familiarity to the environment and colleagues, motivating learner through social contacts between students, experience sharing by the professor, and increase learners' application of the theories through in class and hands on activities.



Figure B.1: Typical modern university classroom settings (Credits: Pinterest ©)

Nevertheless, the learning process in traditional classroom is instructor-centered, time and location constraints, and require dedicated and spacious room to run the course.

As an HCI designer, you are asked to participate in the design team to build a new MOOC classroom system for a fully internet connected college located in a remote area in Sarawak. The professor will be in Kuala Lumpur where most of the teaching and learning, including quizzes will be done remotely. However, the design team is still undecided to determining what would be the most suitable type(s) of interaction and interface to support the new MOOC classroom.

Having done a bit of a background reading on interaction and interface types, experienced e-learning system at UTM, and starting to understand the setting at the new MOOC, you would like to make some recommendations. By focusing on only TWO problems or challenges in hand,

- a. Identify and describe these TWO problems or challenges. (5 marks)
- b. Carefully explain and discuss ONE chosen interaction types for each problem. (4 marks)
- c. Justify TWO chosen interface types related to each problem. (4 marks)
- d. Narrate how both interaction and interface of your chosen types can support and overcome the current problems in the form of scenario. (6 marks)
- e. Support (d) with storyboard to further illustrate the narration. Your storyboard should have the combination of user, scenario, task and the proposed solution(s). (15 marks)

* you may present the answers by problem, i.e. Problem #1 (a-e), Problem #2 (a-e)

Question 2

[33 marks]

One of the world leading oil and gas company is preparing to innovate a new payment system using NFC (Near-Field Communication) technology at self-services cashless petrol pump station. Customer will use their mobile phone with a special app to choose their preference and to confirm payment.

- a. As a user interface designer, you are required to propose user interface (UI) for the best customers' user experience. Design and sketch the propose UI (3-5 screens) for both the app and the pump station. What design principles do you apply?

(12 marks)

- b. The app should give option for new and returning customer with or without loyalty program. Suggest an interaction type between the app of the user with the pump station and justify the reason.

(5 marks)

- c. You need to also consider all user interaction and interface for all level of users (beginner, intermediate and expert). How do you illustrate this in your UI design? Discuss this based on the sketches in (a), or, you may sketch additional interface(s) here.

(5 marks)

- d. Besides using the NFC, customer should be allowed to use other conventional cashless payment method. What metaphor or analogy can you suggest to be considered in the UI of the application? Sketch them.

(5 marks)

- e. Suggest an experiment setup to evaluate the usability of your proposed application design. Mention the type of evaluation, subjects involved and measurement(s) taken.

(6 marks)

[[Ends]]